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Digital Twin-Driven Adaptive Manufacturing Selection for Multiple Part Families in Reconfigurable Flow Lines

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Abstract: Background: Reconfigurable manufacturing systems (RMSs) enable rapid adaptation of machines, tools, and control software to produce different part families. Objectives: This paper addressed a Multiple Part Family Production (MPFP) problem selecting the best reconfiguration policy (RP) when switching production between part families. RPs are represented as sequences of reconfiguration steps and evaluated by number of setups, total reconfiguration cost, and total reconfiguration time. This work presented a conceptual digital twin (DT) based framework for forming RP applicable to reconfigurable flow line (RFL). Method: This framework describes how a DT can model candidate reconfiguration sequences by using real or historical data to estimate time and cost, and support virtually in what if analyses for policy comparison prior to physical changes. The manuscript focuses on framework design, required data flows, and decision logic rather than on implementation or experimental deployment. Results: This work provides a structured DT architecture which acts as reconfiguration decision making tool for engineering managers and a formal representation of reconfiguration policies and evaluation metrics, and a roadmap for integrating data sources, simulation components, and validation procedures. A worked example is also presented to demonstrate how the proposed framework supports a management decision. Illustrative results show that RP_1 and RP_3 require only one reconfiguration, whereas RP_2 incurs an additional change at Stage 3. Consequently, RP_1 is the most efficient policy, followed by RP_3, while RP_2 imposes the highest reconfiguration burden. Conclusion: This work proposes a novel DT-driven RP for MPFP, applicable to an RFL, integrating real-time data analytics into RMS to manage reconfigurable machine tool (RMT) configurations across four production stages. The selected RP is intended to improve delivery performance, capacity utilization, production continuity, and system availability by minimizing setups and reconfiguration time/cost, thereby reducing downtime and maintaining a smoother production flow.

Keywords: Reconfigurable manufacturing system; Digital twin; Reconfiguration policy selection; Reconfigurable machine tool; Changeover optimization; Engineering decision support

1 Introduction

In the current era of industrial transformation driven by Industry 4.0, manufacturing organizations are under immense pressure to continuously innovate, adapt, and respond to unpredictable changes in product demand and customer expectations. The intense global competition and shrinking product life cycles compel manufacturers to adopt systems that can reconfigure rapidly with minimal cost and disruption [1]. Traditional production systems designed for long production runs and stable product lines lack the flexibility required to cope with these new operational realities. Consequently, a reconfigurable manufacturing system (RMS) has emerged as a modern production paradigm capable of balancing high productivity with flexibility and responsiveness [2].

Unlike dedicated or flexible manufacturing systems, an RMS is explicitly designed for cost-effective reconfiguration both in terms of hardware and software allowing manufacturers to produce a variety of product families using modular and scalable configurations [2]. The core philosophy is that a reconfigurable system can swiftly alter its production capacity and functionality, maintaining efficiency under varying product mixes and volumes.

1.1 The Role of Reconfiguration in Multi-Part Family Production

A part family represents a group of products sharing similar design or manufacturing features [3]. In a reconfigurable production environment, these families are produced sequentially across a reconfigurable flow line (RFL), which consists of several stages equipped with reconfigurable machine tools (RMTs) capable of performing multiple operations after minimal setup changes [4]. However, when switching from one part family to another, a part of flow line or the entire one must be reconfigured to align with the operational requirements of the next family. This changeover entails setup time, cost, and resource utilization that directly influence overall production efficiency [5].

Thus, determining an optimal reconfiguration policy (RP) that defines the sequence and method of switching between part families becomes a crucial decision-making problem. The goal is to minimize reconfiguration time, reduce cost, and limit the number of setup transitions without compromising the system's responsiveness or production throughput. Inefficient sequencing can lead to unnecessary setup operations, higher energy consumption, and idle resource time, ultimately increasing total operational cost.

1.2 Need for Adaptive Reconfiguration-Policy Selection

Traditional heuristic or deterministic scheduling models often fail to adapt to the stochastic nature of RMS environments where demand fluctuations and configuration dependencies are frequent. Adaptive algorithms can continuously evaluate real-time parameters such as reconfiguration effort, operation compatibility, and setup interdependencies to update the RP for each required transition within a predetermined part-family production order [6]. These algorithms employ feedback-driven decision mechanisms, making them more suitable for environments with variable product mix, uncertain demand, and limited setup data.

The present study addresses this critical research gap by introducing an adaptive algorithm for Multiple part family production (MPFP) in RFL. The algorithm integrates a set of adaptive decision rules to select the best RP that minimizes the total reconfiguration burden quantified through a multi-objective performance function combining reconfiguration count, reconfiguration cost, and reconfiguration time. The algorithm is termed adaptive because it updates the RP rather than the production sequence in response to changes in demand, shop-floor disturbances, or updated performance criteria. It involves the decision rules that re-evaluate reconfiguration count, reconfiguration cost, and reconfiguration time whenever new data or disruptions occur, ensuring that the selected RP remains suitable under changing operating conditions. Additionally, the integration with digital twin (DT) enables autonomous monitoring and real-time adjustment, allowing the system to continuously adapt to environmental and demand variations while minimizing the overall reconfiguration burden.

1.3 Part Family Formation and Sequencing in Reconfigurable Manufacturing Systems

An RMS is typically composed of RMTs arranged in modular configurations capable of performing multiple operations with minor physical or software-level adjustments. The ability to add, remove, or rearrange modules provides significant competitive advantage in environments characterized by high product variety and variable demand. Several studies have demonstrated that RMSs improve system responsiveness, reduce production lead times, and enhance resource utilization [7, 8].

The foundation of RMS planning lies in the part family formation (PFF) process of grouping components with similar manufacturing characteristics [9]. Once part families are identified, the production of multiple families on an RFL requires systematic sequencing to minimize reconfiguration overheads. Each part family demands a specific sequence of operations across multiple RMTs. When switching from one family to another, setup adjustments including tool changes, fixture reconfigurations, and control parameter modifications are required [10]. Researchers have addressed the sequencing and reconfiguration problem in RMS from different perspectives. Moghaddam et al. [11] proposed a configuration design model for scalable RMS considering variable product demand over its life cycle. Huang et al. [12] proposed a reconfiguration-aware similarity coefficient for grouping RMS parts into families through hierarchical clustering, while Xia et al. [13] developed a maintenance-oriented RP integrating system reliability and cost trade-offs. These works emphasize the importance of timely reconfiguration and sequence optimization to ensure cost-effective transitions between product variants.

1.4 Fundamental Concept of Digital Twin

DTs represent a cutting-edge data-driven approach combining real-time analytics, optimization, and simulation. A DT includes two main parts: a physical component and a virtual counterpart linked through real-time data link, with the virtual part holding detailed information about the physical component manufactured. DTs process vast amounts of real-time and offline data to improve system transparency. This enables extensive information usage for monitoring, predicting, diagnosing, simulating, and optimizing [14]. Some of the features of DT are presented as below:

1.4.1 Definition

A virtual representation of a physical asset, system, or process that updates and interacts with real-world data in real time to optimize performance and decision-making.

• Components: Physical asset/data sources; Virtual model; Data integration layer; Analytics & artificial intelligence (AI).

• Capabilities: Simulation & forecasting; Performance optimization; Remote monitoring; Lifecycle management.

• Services: Asset performance management; Process optimization; Anomaly detection; Simulation & data visualization.

• Applications: Tracking system condition; Generating predictions; Diagnosis; Simulation and optimization.

1.5 Objectives and Research Contribution

This research aims to support production/operations and engineering managers in selecting and scheduling reconfiguration policies for MPFP. The proposed DT evaluates feasible RMT configurations and part family-transition sequences by jointly considering setup frequency, reconfiguration time, reconfiguration cost, production capacity, and operational constraints, thereby supporting production planning, resource allocation, downtime control, and continuity of operations. The specific objective of this research is to design a computationally efficient adaptive sequencing approach for DT-driven MPFP in RMS environments that:

• Models cost, time, and frequency interdependencies.

• Identifies the optimal RP for each transition within a fixed part-family production sequence, ensuring minimal reconfiguration effort.

• Improves responsiveness and ensures consistent throughput during changeovers.

The principal contribution is a DT-based managerial decision-support framework that converts real-time and historical shop-floor data into recommendations for selecting, timing, and authorizing partial or complete reconfiguration actions. An illustrative example is presented in section 5 to exhibit how the projected framework supports a management decision. It enables managers to compare alternative policies before implementation and allocate RMTs, labor, tools, fixtures, and production time while balancing cost, capacity, changeover downtime, delivery requirements, and uninterrupted flow.

1.6 Related Works

The presented body of work collectively demonstrates that integrating DT technology with RMS substantially advances the capability to plan, evaluate, and execute reconfiguration for MPFP while explicitly accounting for reconfiguration cost and time. Key themes emerging from the reviewed works are (1) real-time mapping and model fidelity; (2) optimization and decision-making under disturbance; (3) decentralized and learning-enabled reconfiguration; and (4) explicit treatment of cost/time trade-offs and rapid virtual validation.

1.6.1 Real-time mapping and model fidelity

High-fidelity, real-time DTs are foundational for effective reconfiguration management. Several studies emphasize the need to represent dynamic station and line behavior accurately in virtual space to enable reliable planning and execution [15–19]. Accurate mapping of reconfigurable stations and machines into the DT permits virtual prototyping, parallel virtual–real control, and predictive simulation of multiple product variants [15, 18, 20]. Maintaining and improving model fidelity through incremental learning on process data further strengthens reconfiguration decision quality and reliability [19].

1.6.2 Optimization and decision-making under disturbance

In order to improve throughput and performance in the face of disturbances, it is done by generating near-instantaneous reconfiguration plans. Frameworks combining DT monitoring with rapid configuration optimization have demonstrated the ability to recover throughput losses and to search large configuration spaces quickly [16]. High-speed simulation coupled with capability-based resource models supports rapid evaluation of candidate configurations and selection under time constraints [16]. Analytical and optimization approaches, such as analytical target cascading, multi-objective cuckoo algorithms were applied to jointly handle order scheduling, balancing, and buffer sizing alongside reconfiguration decisions [18, 21]. In this work, the reconfiguration decision is supposed to be made by the production/operations manager, supported by the proposed DT-based decision framework. It is primarily a tactical decision, because it concerns selecting and scheduling reconfiguration policies between part families, while being informed by operational data and constraints.

1.6.3 Decentralized, multi-agent, and learning-enabled approaches

Decentralized and learning-based controllers provide scalable alternatives to centralized planning when layout, agent heterogeneity, or dynamic tasks complicate the reconfiguration. Multi-agent cooperative swarm learning and decentralized decision architectures reported reduce layout compactness, processing time, and rearrangement

cost [22]. Reinforcement learning methods applied through DT-driven environments enable online discovery of reconfiguration policies that incorporate moving, processing, and reconfiguration costs into the reward structure [23]. Combining continuous model updates through learning with reconfiguration management yields progressively improved planning outcomes [19].

1.6.4 Explicit modeling of reconfiguration cost and time

Other works explicitly modeled reconfiguration cost and reconfiguration time as optimization objectives or constraints, reflecting their practical importance in MPFP [18, 21, 23]. Treating reconfiguration holistically integrating physical reconfiguration effort, moving costs, and processing impacts enables generation of minimal or cost-aware reconfiguration solutions and supports trade-off analysis between immediate performance gains and reconfiguration overhead [17, 21]. Predictive estimation of reconfiguration time points i.e., scheduling when to reconfigure further reduces completion times and overall reconfiguration cost in production planning [18].

1.6.5 Rapid virtual validation and variant design

The DT facilitates virtual validation of layout and variant designs prior to physical changes, reducing waste and accelerating changeovers [15, 20–22]. Use cases in electronics and smart-phone assembly illustrate the DT's role in enabling fast variant design iterations, virtual verification of feasibility, and quantification of expected performance or cost impacts [20, 21]. High-fidelity 3D representations and fast simulation support both layout optimization and detailed evaluation of reconfiguration costs for MPFP [22].

1.6.6 Synthesis and gaps

Few past studies proposed an evolutionary ecosystem in which DTs act as the central enabler linking monitoring, simulation, optimization, and learning for RMS reconfiguration. Effective systems combine real-time DT mapping, capability-aware resource models, rapid simulation, and optimization algorithms that explicitly account for reconfiguration cost and time, while leveraging decentralized and learning-based methods for scalability and adaptability. Remaining challenges suggested by the works include standardized metrics for trade-off evaluation across cost/time/performance dimensions, integration of human factors during semi-automatic reconfiguration, and systematic approaches to guarantee DT model validity over long-term deployments issues. Future works should address to make reconfiguration solutions robust in large-scale MPFP environments [17, 19].

1.7 Literature Summary

This section presents a research summary (Table 1) of some of the most relevant works, which involve manufacturing setting, decision criteria, data type, online or offline operation, treatment of multiple part families.

Table 1. Literature summary

Ref.	Manufacturing Setting	Decision Criteria	Data Type	Online or Offline Operation	Treatment of Multiple Part Families
[5]	RFL producing multiple part families; RMS-based environment with configurable stations	Selection of optimal part-family manufacturing sequence based on reconfiguration effort, setup changes, and processing feasibility. Algorithm evaluated alternative sequences to minimize total reconfiguration burden	Operation-sequence data for each part family, machine capability data, reconfiguration requirements; algorithmic evaluation data (rule-based + computational)	Offline decision support: algorithm computes optimal sequence before production; not a real-time DT-synchronized system	Central focus: the algorithm explicitly handles multiple part families and selects the best manufacturing order to minimize reconfiguration
[15]	Mixed-model flow line with reconfigurable stations	Minimize setup/reconfiguration time using dual DT replicas	Real-time product-arrival data; DT station state	Online (incoming twin prepares ahead of arrival)	Yes: mixed-model flow line inherently handles multiple variants
[16]	Manufacturing line with disturbances	Throughput recovery; optimal reconfiguration under time-change disturbances	DT monitoring data; capability ontology; simulation data	Online (DT triggers reconfiguration)	Not explicitly multi-family; disturbance-driven
[17]	Generic RMS	Holistic reconfiguration management; completeness vs. 32 DT requirements	DT models; system-level configuration data	Both (framework supports continuous DT synchronization)	Not explicitly multi-family
[18]	RMS with machine tools + production line	Minimize part completion time + reconfiguration cost; predict reconfiguration time point	DT model + simulation samples; multi-objective cuckoo search	Offline optimization + online prediction	Yes: optimization considers part-completion time across variants

[19]	CPPS	Find stable/optimal configurations via adaptive process models	Intelligent DT + adaptive process models; process data	Online self-organized reconfiguration	Not explicitly multi-family
[20]	3C electronic product assembly line	Variant design; rapid reconfiguration; avoid cost waste	DT virtual–real linkage; simulation of assembly line	Offline design + online validation	Yes: variant design explicitly supports multiple product families
[21]	Semi-automatic electronic assembly line with frequent changeovers	Minimal reconfiguration (order scheduling, line balancing, buffer allocation)	DT mirror; GKE knowledge encapsulation; max-plus semiring evaluation	Offline optimization + online synchronization	Yes: frequent changeovers imply multi-family production
[22]	Reconfigurable robotic assembly cells (aerospace cases)	Layout compactness, rearrangement cost, production time	3D-DT facility models; multi-agent swarm learning	Online learning in DT environment	Yes: aerospace cases involve multiple product variants
[23]	Smart manufacturing system with 5 RMTs producing 4 parts	Maximize return via DRL (cost, moving cost, processing cost)	DT-driven state/action space; RMT process data	Online (Deep Q-Network-based RP search)	Yes: explicitly 4 different parts (multi-family)

Note: 3C = computer, communication, and consumer electronics; CPPS = cyber-physical production systems; DRL = deep reinforcement learning; DT = digital twin; GKE = generalized knowledge encapsulation; RFL = reconfigurable flow line; RMS = reconfigurable manufacturing system; RMT = reconfigurable machine tool.

1.8 Research Gaps

Despite considerable progress, several research gaps remain unresolved in the area of RMS sequencing and reconfiguration planning:

1. Limited consideration of multi-objective dependencies: Most previous studies optimize either reconfiguration cost or time, but rarely integrate both along with setup frequency in a unified decision model.

2. Static reconfiguration-policy assumptions: Many models evaluate reconfiguration alternatives using fixed input data and do not update policy selection when the system state changes.

3. Inadequate scalability: Existing algorithms often struggle to handle high part family diversity or stochastic variations in real-world production lines.

4. Lack of integration with DT: The conventional approaches are not designed to operate with DT frameworks that demand real-time adaptation and self-learning capabilities.

Consequently, there is a need for a DT-based decision-support procedure that can continuously assess and update reconfiguration policies within a predetermined part-family production order by considering changing operating conditions and multiple performance measures.

The paper addresses a multi-criteria engineering management challenge of selecting RPs for MPFP in an RFL, requiring managers to balance reconfiguration time, cost, and number of set-ups while meeting throughput and technological constraints. The study further shows how a DT-driven decision process supports this selection by enabling systematic planning, comparison, and governance of reconfiguration actions for successive part-family transitions under diverse performance criteria and operational constraints.

The remainder of the paper is structured into several sections i.e., Section 2 focuses on the methodology; Section 3 presents the framework for DT-driven RP for MPFP; Section 4 presents the Mathematical formulation; Section 5 presents an illustration for demonstrating managerial decision support; Section 6 include discussion, while Section 7 concludes with major findings and Section 8 provides limitations of the work and directions for future research.

2 Methodology

An illustrative case can be drawn from an RFL designed under the principles of RMS, which enables the sequential production of multiple part families. Consider a manufacturing enterprise producing several families of parts i.e., $f_1, f_2, f_3 \ldots f_n$ on the RFL in succession. Each part family is characterized by distinct operational requirements, designated as '1' in Table 2, necessitating reconfiguration of the line prior to initiating production. While one family is being processed, the remaining families await their turn. Owing to the variability in operating requirements, the RMT setups must be adjusted at each stage before commencing production of the subsequent family. Production can only begin once the RFL has been reconfigured to meet the specifications of the next family, and this iterative cycle continues until the final family has been completed.

The proposed conceptual framework is designed to be system-agnostic and adaptable, focusing on the algorithmic architecture and decision-making logic rather than detailed numerical case studies or a deployed, data-driven implementation.

A fair benchmarking requires realistic shop-floor inputs and identical constraints across policies; using arbitrary or synthetic values would risk producing case-specific or misleading conclusions. Therefore, formal empirical comparisons are deferred to a planned follow-up implementation study that will populate the framework with real or validated historical data and perform controlled comparisons and sensitivity analyses. This approach ensures broad applicability across diverse manufacturing environments, with numerical validations to be conducted once specific shop-floor data are available. By prioritizing framework development over case-specific computations, the study maintains its conceptual integrity and generalizability. Hence, Table 1 presents literature summary; Table 2, Table 3, Table 4, Table 5, and Table 6 are intended to illustrate the structural inputs and decision variables of the proposed adaptive sequencing method and Table 7, Table 8, and Table 9 present the findings.

Table 2. Required operations (O_l) and sequences for producing the product families (f_j)

$O_l \rightarrow f_j \downarrow$	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
f_1	0	1	1	1	0	1	0	0
f_2	1	0	1	0	1	0	1	0
f_3	1	0	0	0	0	1	1	1
f_4	0	1	0	0	1	0	0	1
f_5	0	0	1	1	0	1	0	0

Table 3. Production rates $\left(R_{w_i}^{O_l}\right)$ of reconfigurable machine tool configurations (T_k^p) for executing operations (O_l)

$O_l \rightarrow T_k^p \downarrow$	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
T_1^1	14	–	16	–	18	–	–	–
T_1^2	22	20	10	17	–	–	12	–
T_2^1	–	–	20	–	–	13	–	16
T_3^1	10	15	–	–	–	–	18	–
T_3^2	14	–	–	15	–	–	17	–
T_4^1	9	12	15	–	–	10	–	20
T_4^2	–	–	–	–	16	–	–	19

Note: – indicates that the corresponding operation cannot be performed by the machine tool configuration.

Table 4. Reconfiguration time and reconfiguration cost incurred in configuration changeover ($l_{k\rightarrow k'}^{p\rightarrow p'}$,$c_{k\rightarrow k'}^{p\rightarrow p'}$)

T_k^p	T_1^1	T_1^2	T_2^1	T_3^1	T_3^2	T_4^1	T_4^2
T_1^1	(0,0)	(30,10)	(50,35)	(40,20)	(80,40)	(90,30)	(60,45)
T_1^2	(30,10)	(0,0)	(80,20)	(120,50)	(100,25)	(90,40)	(80,50)
T_2^1	(50,35)	(80,20)	(0,0)	(60,40)	(90,30)	(50,35)	(120,50)
T_3^1	(40,20)	(120,50)	(60,40)	(0,0)	(80,50)	(80,20)	(50,35)
T_3^2	(80,40)	(100,25)	(90,30)	(80,50)	(0,0)	(50,35)	(90,30)
T_4^1	(90,30)	(90,40)	(50,35)	(80,20)	(50,35)	(0,0)	(90,40)
T_4^2	(60,45)	(80,50)	(120,50)	(50,35)	(90,30)	(90,40)	(0,0)

Table 5. Potential reconfigurable machine tool configurations(T_k^p) with operations (O_l) performed on part families (f_j)

$O_l \rightarrow f_j \downarrow$	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
f_1	–	T_1^2, T_3^1, T_4^1	T_1^1, T_1^2, T_4^1	T_1^2,T_3^2	–	T_2^1, T_4^1	–	–
f_2	T_1^1,T_3^2	–	T_1^1,T_2^1,T_4^1	–	T_1^1,T_4^2	–	T_1^2,T_3^1,T_3^2	–
f_3	T_1^2,T_3^1,T_4^1	–	–	–	–	T_2^1,T_4^1	T_1^2,T_3^1,T_3^2	T_2^1,T_4^1,T_4^2
f_4	–	T_1^2,T_3^1,T_4^1	–	–	T_1^1, T_4^2	–	–	T_2^1,T_4^1,T_4^2
f_5	–	–	T_1^1,T_2^1,T_4^1	T_1^2,T_3^2	–	T_2^1,T_4^1	–	–

Note: – indicates that no reconfigurable machine tool configuration is listed for the corresponding part family and operation.

Table 6. Feasibility mapping

Stage No.	Operation Transition in $f_2 \rightarrow f_3$	Feasible Configurations for f_2 (Table 5)	Feasible Configurations for f_3 (Table 5)	Configuration Overlap and Changeover Requirement
Stage 1	$O_1 \rightarrow O_1$	T_1^1, T_3^2	T_1^2, T_3^1, T_4^1	No overlap; changeover required
Stage 2	$O_3 \rightarrow O_6$	T_1^1, T_2^1, T_4^1	T_2^1, T_4^1	Overlap: T_1^2, T_4^1; no changeover required
Stage 3	$O_5 \rightarrow O_8$	T_1^1, T_4^2	T_2^1, T_4^1, T_4^2	Overlap: T_4^2; no changeover required
Stage 4	$O_7 \rightarrow O_7$	T_1^2, T_3^1, T_3^2	T_1^2, T_3^1, T_3^2	All configurations overlap; no changeover required

Table 7. Reconfiguration policy 1—lowest reconfiguration cost at each stage

Stage/Operation	Configuration for f_2	Configuration for f_3	Configuration Changeover	Time (min)	Cost (10^3 USD)
Stage 1 ($O_1 \rightarrow O_1$)	T_1^1	T_1^2	$T_1^1 \rightarrow T_1^2$	30	10
Stage 2 ($O_3 \rightarrow O_6$)	T_2^1	T_2^1	No changeover required	0	0
Stage 3 ($O_5 \rightarrow O_8$)	T_4^2	T_4^2	No changeover required	0	0
Stage 4 ($O_7 \rightarrow O_7$)	T_3^1	T_3^1	No changeover required	0	0
		TOTAL		30	10

Table 8. Reconfiguration policy 2—highest production-rate configuration at each stage

Stage/Operation	Configuration for f_2	Configuration for f_3	Configuration Changeover	Time (min)	Cost (10^3 USD)
Stage 1 ($O_1 \rightarrow O_1$)	T_1^1	T_1^2	$T_1^1 \rightarrow T_1^2$	30	10
Stage 2 ($O_3 \rightarrow O_6$)	T_2^1	T_2^1	No changeover required	0	0
Stage 3 ($O_5 \rightarrow O_8$)	T_1^1	T_4^1	$T_1^1 \rightarrow T_4^1$	90	30
Stage 4 ($O_7 \rightarrow O_7$)	T_3^1	T_3^1	No changeover required	0	0
		TOTAL		120	40

Table 9. Reconfiguration policy 3—fewest configuration changes

Stage/Operation	Configuration for f_2	Configuration for f_3	Configuration Changeover	Time (min)	Cost (10^3 USD)
Stage 1 ($O_1 \rightarrow O_1$)	T_1^1	T_3^1	$T_1^1 \rightarrow T_3^1$	40	20
Stage 2 ($O_3 \rightarrow O_6$)	T_2^1	T_2^1	No changeover required	0	0
Stage 3 ($O_5 \rightarrow O_8$)	T_4^2	T_4^2	No changeover required	0	0
Stage 4 ($O_7 \rightarrow O_7$)	T_3^1	T_3^1	No changeover required	0	0
		TOTAL		40	20

2.1 Assumptions

The following assumptions are made in this work:

(i). Deterministic demand for each of the five-part families (which may differ) and a predefined production sequence, while the proposed procedure selects the RP for each required family transition.

(ii). The RFL is modeled as a sequential system in which stages may operate concurrently on different units, subject to sufficient intermediate-buffer capacity; buffer capacities are assumed fixed and non-constraining in the baseline framework.

(iii). To isolate reconfiguration-policy effects, machine failures, unplanned maintenance, and worker-availability constraints are excluded; equipment and personnel are assumed available, with such disruptions identified as extensions for future DT implementation.

(iv). Changing from one configuration to another is bi-directional and thus has the same time and cost in reverse transition.

(v). The order of families is fixed, and adaptation applies only to policy selection under changing conditions.

(vi). Production time is calculated only for capacity and throughput-feasibility checking. It is not included as an optimization criterion in the objective function.

Since multiple RMT configurations may be capable of executing the same operation, it becomes essential to evaluate and select the configuration that minimizes installation and/or reconfiguration cost while also ensuring rapid adaptability. The Reconfiguration path thus defines the sequence of transitions required when shifting from one family's production setup to another. Nomenclature outlines the parameters and notations considered in addressing this problem.

The operational sequence necessary for manufacturing part families is outlined in Table 2. The production rates of RMT configurations (in parts per hour) along with their operational capabilities are presented in Table 3, this data is only used for feasibility and throughput validation, not for optimization. Table 4 specifies the reconfiguration time (min) required to switch between configuration setups and the associated reconfiguration cost (10^3 USD). Finally, Table 5 lists the candidate RMT configurations capable of executing the operations required for processing the part families.

2.2 Introduction of the Digital Twin-Based Multiple Part Family Production Framework in Factory Settings

A manufacturing enterprise introducing the DT-based MPFP framework would collect real-time machine states, tool and fixture identifiers, part-family operation requirements, cycle times, and historical reconfiguration logs. These data streams are pre-processed at the edge (cleaning, mapping, and timestamp synchronization) before being transferred to the Knowledge Database through standard industrial middleware such as Open Platform Communications-Unified Architecture (OPC-UA) or Message Queuing Telemetry Transport (MQTT). The DT uses this processed information to evaluate candidate reconfiguration policies and generate recommendations consistent with the performance criteria of reconfiguration time, reconfiguration cost, and number of setups.

The DT-based MPFP framework involves interaction of DT with shop-floor devices through a data-integration layer that links sensors, RMT controllers, and the Knowledge Database with the virtual simulation environment. A lightweight decision-support interface presents the recommended RP to factory personnel. Decision authority follows a simple governance structure: Operators execute approved changeovers, line supervisors authorize partial reconfigurations, and engineering managers approve full configuration transitions. This workflow ensures that DT-generated recommendations are reviewed and validated before physical implementation.

3 Digital Twin-Driven Reconfiguration Policy for Multiple Part Family Production

The flow chart for creating an RP in MPFP within RMS is shown in Figure 1. The first step is to identify the part families and the order in which they will be processed. Then, it is checked whether all part families have already been produced. If not, the first unfinished part family must be produced. Before production begins, it is verified if a configuration change is needed at the production stage. If a change is required, it should follow these rules:

(a). Keep the number of setup changes as low as possible.

(b). If a set-up change is necessary, it should be done in a way that minimizes the number of reconfigurations, the time taken, and the cost involved. Part families should be produced using the less expensive RMT configurations that can still perform the required operations.

If no configuration change is needed on the RFL, the part family is produced using the current setup. If a change is required, it must follow the two rules mentioned earlier. After the configuration is updated, the part family is produced. Once production of that family is finished, another check is made to see if all part families are completed. This cycle repeats until the last part family has been produced.

Figure 1 depicts physical changeovers which are monitored using digital work instructions, sensor feedback, and stepwise verification checkpoints. Each RMT transition is validated and recorded in the Knowledge Database to refine future DT predictions. If deviations occur, the system triggers a short planned interruption window to

maintain safe and controlled reconfiguration. This monitoring loop ensures that DT-recommended policies translate into reliable on-floor execution.

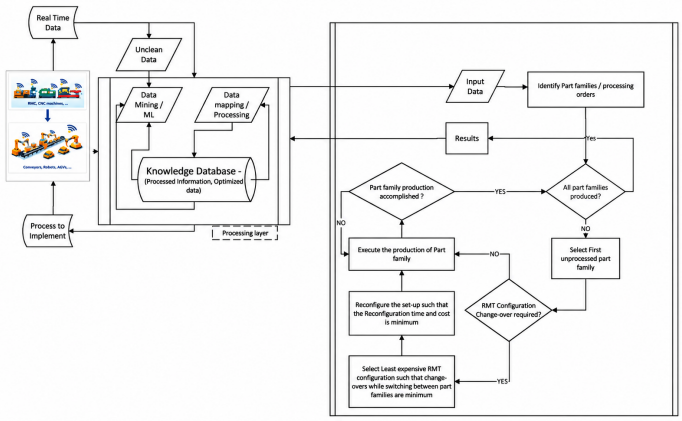


Figure 1. Digital twin-driven reconfiguration policy for multiple part family production

Note: AGVs = automated guided vehicles; CNC = computer numerical control; ML = machine learning; RMC = reconfigurable manufacturing cell; RMT = reconfigurable machine tool.

3.1 Operation Sequences Selection of Multiple Part Family Production Using the Adaptive Algorithm

In an RFL with four production stages, five part families are scheduled for production in a defined order. The adaptive algorithm updates the RP for a fixed part-family order; it does not determine the production sequence. Each family requires specific operations, which differ in type, sequence (Table 2), and the possible RMT configurations that can perform them (Table 5). The same operation may be executed by different RMT configurations, but these vary in production rates (Table 3). Therefore, the most suitable RMT configuration must be selected. Once all the operations on a part family are completed, the next family is prepared for processing. Since the requirements of the subsequent family differ from the previous one, the production stages must be reconfigured accordingly. The choice of reconfiguration time and cost (Table 4) significantly influences the overall effectiveness of the RMS. Thus, when shifting from a part family to another, the two previously stated guidelines must be applied.

Consider a MPFP sequence presented in a generic form $f_j \rightarrow f_{j'} \rightarrow f_{j''} \rightarrow f_{j'''}$, processed across the four stages of the RFL. These stages are equipped with suitable RMT configurations capable of performing certain required operations. A multi-level RP, comprising levels (i) $f_j \rightarrow f_{j'}$, (ii) $f_{j'} \rightarrow f_{j''}$ and (iii) $f_{j''} \rightarrow f_{j'''}$, is applied as per the guidelines outlined earlier. The corresponding minimum reconfiguration time and cost are presented in Table 4. Likewise, reconfiguration policies can be suggested for other MPFP sequences as well.

Part families with similar operation sequences can be produced consecutively with minimal or no changeovers. However, part families with entirely different production requirements demand major setup changes on the RFL, which increases the number of reconfigurations, as well as the associated time and cost.

3.2 Pseudocode for the Framework

The pseudocode for the framework depicted in Figure 1 is presented as below:

Algorithm 1. Pseudocode for the proposed reconfiguration framework

```
// Initialize
INIT KnowledgeDB      // holds processed data and ML models
CONNECT RealTimeStream      // live sensor / shop-floor feed
LOAD Orders, Machines, RMT_Configs

// Background: keep KnowledgeDB updated
FUNCTION UpdateKnowledgeDB():
    LOOP while system_running:
        raw = RealTimeStream.read()
        clean = Clean(raw)
        features = ExtractFeatures(clean)
        preds = KnowledgeDB.Predict(features)
        KnowledgeDB.Store(features, preds)
```

```
    END LOOP
END FUNCTION

// Main short reconfiguration loop
FUNCTION RunReconfiguration():
    part_families = IdentifyPartFamilies(Orders)
    current_state = QueryMachineState()
    schedule = []

    WHILE exists pf in part_families with pf.remaining > 0:
        pf = SelectFirstUnprocessed(part_families)

        IF pf.remaining <= 0:
            mark pf completed
            CONTINUE
        END IF
        candidates = FilterCompatibleRMTs(pf, RMT_Configs, current_state)

        // Choose config that minimizes changeover time + cost
        best = ARGMIN candidates BY
            (WeightedEstimateChangeoverTime(current_state, cfg) + WeightedEstimateChangeoverCost(current_state,
cfg))

        IF current_state.config != best.config:
            plan = PlanReconfiguration(current_state, best)
            ExecuteReconfiguration(plan)      # minimize time & cost
            current_state = best.state
        END IF

        production = CreateProductionPlan(pf, best)
        ExecuteProduction(production)

        WHILE not production.complete:
            telemetry = RealTimeStream.read_short()
            AdjustIfNeeded(production, telemetry, KnowledgeDB)
            IF production.failed:
                HandleFailure(production)
                BREAK
            END IF
        END WHILE

        pf.remaining = pf.remaining - production.completed_qty
        IF pf.remaining == 0:
            pf.complete = TRUE
        END IF
        Append(schedule, {pf.id, best.id, production.metrics})
    END WHILE

    RETURN CompileResults(schedule)
END FUNCTION

// Start system
START UpdateKnowledgeDB() in background
results = RunReconfiguration()

    SAVE results to KnowledgeDB
NOTIFY stakeholders
```

4 Mathematical Formulation

The mathematical model focuses solely on minimizing setup count, reconfiguration time, and reconfiguration cost, which quantify the reconfiguration burden between part-family transitions. In contrast, production-rate and processing-time data serve only as feasibility and throughput-validation parameters, ensuring that selected RMT configurations satisfy capacity requirements. These rate-based checks support decision feasibility but are intentionally excluded from the optimization objective.

• Sets and Indices

$$W = \{w_i \mid i = 1, 2, \ldots, m\} \quad \text{(workstations on RFL)}$$
$$F = \{f_j \mid j = 1, 2, \ldots, 5\} \quad \text{(part families)}$$
$$T = \{T_k \mid k = 1, 2, 3, 4\} \quad \text{(RMTs)}$$
$$T_k = \{T_k^p \mid p = 1, 2\} \quad \text{(configurations of RMT , } T_k\text{)}$$
$$L = \{O_l \mid l = 1, 2, \ldots, 8\} \quad \text{(operations)}$$

• Parameters

q_j = demand (number of parts) of part family, f_j

$\tau_{k\to k'}^{p\to p'}$ = reconfiguration time (min) from configuration T_k^p to $T_{k'}^{p'}$

$C_{k\to k'}^{p\to p'}$ = reconfiguration cost (1000 USD) from T_k^p to $T_{k'}^{p'}$

$R_{w_i}^{O_l}$ = production rate $\left(\dfrac{\text{parts}}{\text{hr}}\right)$ at workstation w_i for operation O_l

Cap_{w_i} = available capacity (hr) at workstation, w_i

$\lambda_t,\ \lambda_c \geq 0$ weights for reconfiguration time and reconfiguration cost in objective function

• Decision Variables

$$x_{jlikp} = \begin{cases} 1, & \text{if operation } O_l^{f_j} \text{ of part family } f_j \text{ is processed at workstation } w_i \\ 0, & \text{otherwise} \end{cases}$$

$$u_{kp} = \begin{cases} 1, & \text{if configuration } T_k^p \text{ is used at least once} \\ 0, & \text{otherwise} \end{cases}$$

$$y_{kk'pp'} = \begin{cases} 1, & \text{if a changeover from } T_k^p \text{ to } T_{k'}^{p'} \text{ occurs} \\ 0, & \text{otherwise} \end{cases}$$

• Objective Function

$$\min\ Z = \lambda_t \sum_{k,k'} \sum_{p,p'} y_{kk'pp'} \tau_{k\to k'}^{p\to p'} + \lambda_c \sum_{k,k'} \sum_{p,p'} y_{kk'pp'} C_{k\to k'}^{p\to p'}$$

• Constraints

(a). Operation assignment (each operation of each part family must be assigned once)

$$\sum_{i\in W} \sum_{k\in T} \sum_{p\in P_k} x_{jlikp} = 1; \quad \forall j \in F,\ \forall l \in L_j$$

(b). Workstation capacity

$$\sum_{j\in F} \sum_{l\in L} \sum_{k\in T} \sum_{p\in P_k} \frac{q_j x_{jlikp}}{R_{w_i}^{O_l}} \leq Cap_{w_i}, \quad \forall i \in W$$

(c). Link between assignment and configuration usage
If any operation uses configuration T_k^p, then $u_{kp} = 1$:

$$u_{kp} \geq x_{jlikp}; \quad \forall j \in F,\ l \in L_j,\ i \in W,\ k \in T,\ p \in P_k$$

(d). Link between configuration usage and possible changeovers
If two configurations are used, a possible changeover is activated:

$$y_{kk'pp'} \geq u_{kp} + u_{k'p'} - 1; \quad \forall k, k' \in T,\ p \in P_k,\ p' \in P_{k'}$$

(e). Binary domains

$$x_{jlikp} \in \{0, 1\}, \quad u_{kp} \in \{0, 1\}, \quad y_{kk'pp'} \in \{0, 1\}$$

5 Illustration for Demonstrating Managerial Decision Support

To illustrate how the proposed framework assists engineering-management decision-making, a small worked example is presented using the operational requirements (Table 2), production-rate capabilities (Table 3), and reconfiguration time/cost matrix (Table 4). The example considers the transition from Part Family $f_2 \rightarrow f_3$, for which three feasible RPs (RP_1, RP_2, RP_3) exist based on the alternative RMT configurations capable of executing the required operations.

5.1 Identify Feasible Configurations for Part Families f_2 and f_3

The RFL used in this study has four production stages. For the $f_2 \rightarrow f_3$ transition, f_2 requires operations O_1, O_3, O_5 and O_7; f_3 requires O_1, O_6, O_7 and O_8 (Table 2). Using Table 3, the feasible RMT configurations for these operations are extracted. Table 5 was used to determine which configurations may serve f_2' s operation and which may serve f_3's operation for every stage. Matching operation-to-operation across the two families for four stages: Stage 1 ($O_1 \rightarrow O_1$), Stage 2 ($O_3 \rightarrow O_6$), Stage 3 ($O_5 \rightarrow O_8$) and Stage 4 ($O_7 \rightarrow O_7$) is presented in Table 6, which shows this feasibility check as well as whether a zero-cost "no-change" option exists at that stage or not.

5.2 Candidate RPs and Computation of Reconfiguration Count, Time, and Cost

Using the feasibility mapping (Table 6), Rows with identical configurations for f_2 and f_3 are marked "no changeover required" with zero reconfiguration time and cost. At Stage 1, there is no overlap between f_2' s feasible set $\{T_1^1, T_3^2\}$ and f_3' s feasible set $\{T_1^2, T_3^1, T_4^1\}$, so a change is unavoidable; among all feasible pairs, $T_1^1 \rightarrow T_1^2$ gives the lowest cost (10, at 30 min) from Table 4. On Stages 2–4, the feasible sets overlap i.e., $\{T_2^1, T_4^1\}$ for Stage 2, T_4^2 for Stage 3, and all set of configurations for Stage 4, so the zero-cost "no changeover required" option is selected, consistent with RP_1 rule of minimizing cost at every stage. The required stage-by-stage tables for RP_1, RP_2 and RP_3 are mentioned in Table 7, Table 8, Table 9. The RPs are defined as RP_1: Select the lowest-cost configuration for each operation; RP_2: Select the fastest-rate configuration for each operation; RP_3: Select configurations that minimize the number of configuration changes between f_2 and f_3. Using Table 4, the reconfiguration count and transition cost/time between configurations is computed and the results are presented in Table 7, Table 8, Table 9. Hence, the stage-wise tables for RP_1, RP_2 and RP_3 are presented in Table 7, Table 8, Table 9 respectively.

5.3 Managerial Interpretation of Trade-Offs

A production or engineering manager evaluates the three policies i.e., RP_1, RP_2 and RP_3 by balancing changeover burden and operational continuity. The result in the Table 7, Table 8, and Table 9 shows that RP_1 and RP_3 both require only a single reconfiguration (Stage 1) because Stages 2–4 admit a zero-cost "no changeover required" option, whereas the rate-maximizing rule of RP_2 forces an additional costly change at Stage 3. RP_1 is therefore the cheapest and fastest policy, RP_3 is a close second, and RP_2 remains clearly the most burdensome policy.

6 Discussion

The proposed conceptual DT-based framework for selecting RPs in MPFP builds on existing research showing how DTs can support decision-making in RMS. Additionally, the selected RPs are not only a machine-configuration solution; selecting them is a tactical production-management decision that determines when resources are withdrawn, reassigned, or returned to production during family changeovers. By minimizing unnecessary setups and balancing reconfiguration time and cost against production requirements, the framework can support capacity utilization, delivery reliability, downtime reduction, and continuity of operations; however, these benefits remain to be empirically validated because the present study is conceptual and has not yet been deployed or experimentally benchmarked.

The framework integrates the system by linking RMT configurations with real-time DT data flows, allowing configuration states, operation requirements, and transition costs to be continuously synchronized. These data streams feed a managerial decision layer that evaluates feasible reconfiguration policies against operational objectives such as throughput, downtime, and cost. Together, physical resources, information flows, and decision authority operate as a single coordinated mechanism for selecting and governing reconfiguration actions.

By presenting an illustrative example in Section 5, the framework provides clearer planning and authorization workflows by structuring how managers evaluate and approve reconfiguration actions. Its coordinated use of DT data reduces unnecessary downtime through timely, evidence-based configuration updates. By formalizing the comparison of alternative policies, it enables a consistent and transparent evaluation process that strengthens operational decision-making.

In relation to this work, another work [15] developed a DT approach aimed at reducing setup times in mixed-model flow lines. The proposed DT evaluated different station configurations and recommends the sequence of changes that minimizes changeover time and the resulting downtime for operators. Their findings show that using a DT to test different setup options in advance can noticeably reduce the time needed for actual physical reconfiguration. This supports the idea that reconfiguration policies expressed as sequences of reconfiguration steps should first be

evaluated in the DT environment before being applied on the real system, using metrics such as number of setups, total reconfiguration time, and total reconfiguration cost [15].

Fu et al. [16] expanded this concept by applying DTs to handle unexpected disturbances in smart manufacturing. Their DT framework continuously monitors the production line for disruptions; creates alternative line configurations using an optimization module; and quickly tests these options through simulation. This approach helped in recovering up to 63% of lost throughput during disturbance events. Results show that DT-driven reconfiguration can keep production running and maintain system availability even when conditions change. Their findings support the inference of the present work i.e., the chosen RP should improve delivery performance, capacity use, and production continuity by reducing setups and reconfiguration time and cost. However, while Fu et al. focused on reconfiguration triggered by disturbances, the current study generalizes the decision problem to MPFP, where reconfiguration is planned in advance when switching between part families rather than caused by unexpected events.

Another DT framework specifically for reconfiguration management, emphasized a conceptual architecture and evaluation rather than full-scale deployment [17]. Their framework defined modules for data acquisition, model management, reconfiguration planning, and evaluation, and demonstrated through case studies how DT-based reconfiguration management can systematically compare alternative policies. This research is closely aligned with the contribution of present work of a structured DT architecture that acts as a reconfiguration decision-making tool for engineering managers, and a formal representation of RPs and evaluation metrics [17]. A conceptual focus is linked with the present study that it provides roadmaps for integrating data sources, simulation components, and validation procedures, rather than detailed implementation of algorithms or hardware integration, reinforcing the legitimacy of a conceptual, architecture-oriented contribution.

A DT-based method was developed to predict reconfiguration time points in RMS [18]. Their approach used DT models to forecast when reconfiguration should occur and how long it will take, thereby reducing uncertainty in planning and scheduling. The reported results showed alignment between planned and actual reconfiguration times, which directly supported this study's emphasis on using real or historical data within the DT to estimate reconfiguration time and to compare candidate RPs on this basis [18].

A DT with self-organized reconfiguration management, built on an adaptive process model [19], showed why DTs need to be flexible and able to update themselves as conditions change. Such adaptive, feedback-driven DTs can revise their internal process models and make reconfiguration decisions automatically. This supported the idea that a DT-based reconfiguration-policy framework should not only store fixed reconfiguration sequences, but also adjust how it evaluates those sequences whenever new data such as updated processing times or failure rates becomes available.

Another DT-based method [20] designed and evaluated different configuration and layout variants for an assembly line producing computer, communication, and consumer electronics (3C) products. Their DT models allow rapid testing of design options before making physical changes, which reported shortened development time and improved decisions, especially when many product variants must be handled. This idea is similar to the MPFP context, where multiple product families require frequent reconfiguration. Their work is in line with the view that DTs can model and compare candidate reconfiguration sequences using historical and real-time data, enabling "what-if" analysis before any physical adjustments are made.

Zhang et al. focused on DT-enabled optimal reconfiguration of a semi-automatic electronic assembly line with frequent changeovers, explicitly optimizing reconfiguration decisions to handle high changeover frequency [21]. Their results demonstrate improvements in throughput and reductions in downtime when DT-based optimization is used to select reconfiguration strategies, particularly under frequent product switches. This directly reinforces the manuscript's emphasis on minimizing setups and reconfiguration time/cost in MPFP, as frequent part-family changes can otherwise lead to excessive downtime and reduced capacity utilization [21]. Zhang et al. provided empirical evidence that DT-enabled optimization of reconfiguration policies yields tangible operational benefits, supporting the manuscript's intended impact on delivery performance and production continuity.

A dynamic layout optimization method of reconfigurable robotic assembly cells used DT models as the environment in which agents learn and evaluate layout changes [22]. Their results showed that multi-agent learning can discover dynamic layouts that improve performance metrics. Although their focus was on layout optimization rather than explicit RP sequences, the underlying principle that DTs can host learning-based optimization of reconfiguration decisions extends this conceptual framework. It suggests that the proposed DT architecture for RP evaluation could be augmented with multi-agent or swarm-based learning modules to automatically generate and refine RPs for MPFP in RFL [22].

Huang et al. proposed a DRL-based dynamic reconfiguration planning approach for DT-driven smart manufacturing systems with RMTs [23]. Their framework integrated process planning, reconfiguration, and scheduling, and uses DRL to learn policies that balance reconfiguration cost, time, and production performance. The reported results indicate that DRL-based policies can outperform heuristic baselines in terms of throughput and responsiveness to changing demands [23]. The study demonstrated that DT-driven reconfiguration planning can be coupled with

advanced learning algorithms. This provided a clear future direction for extending the proposed RP framework from rule-based or heuristic evaluation toward learning-based policy optimization in MPFP contexts [23].

In this context, the proposed framework can be viewed as a synthesis and extension of prior DT-driven RMS research works, adopts the proven benefits of DT-based reconfiguration decision support [15–17, 21], incorporates predictive and evaluative metrics aligned with time and cost considerations [18, 21], and positions itself to integrate emerging optimization and learning approaches for dynamic, data-driven RP selection in future work [19, 22, 23]. The manuscript addresses a practically important yet under-specified decision problem in RMS, and its DT-driven RP framework is well grounded in, and meaningfully connected to, the core results of the past studies [15–23].

7 Conclusions

This work presents a novel methodology for MPFP on an RFL by integrating a DT-driven RP with the RMS. Multiple, dissimilar part families each requiring distinct operations and operation sequences are processed across the four production stages of the RFL. When reconfiguration is necessary, existing RMT configurations on the stages are converted to new configurations between the completion and start of the next part family. For a predetermined part-family production order, the RP for each required transition is evaluated using the number of reconfiguration setups, reconfiguration time, and reconfiguration cost.

The DT-driven RP is proposed to continuously collect and process real-time shop-floor data into a Knowledge Database (data cleaning, mapping, data-mining and machine learning). This data-driven decision loop is developed to perform a predictive assessment of whether an RMT changeover is required, quantify trade-offs among alternative RMT configurations (production rates, installation/removal costs), and select the configuration that minimizes changeovers, time, and cost while meeting operational requirements. For each pair of consecutive part families in the predetermined production order, the DT-based procedure compares the feasible reconfiguration policies and identifies the policy with the lowest reconfiguration burden; part families with distinct requirements trigger partial or full reconfigurations, with the DT-monitoring execution and updating the Knowledge Database to refine future decisions.

The proposed work is designed to improve responsiveness, reduce needless reconfigurations, and may support informed scheduling decisions under realistic, data-driven constraints and can support managers in evaluating trade-offs among alternative RPs and selecting the most suitable choices for MPFP. Three RPs are evaluated in Table 7, Table 8, and Table 9. The selected RPs supports tactical production management by optimizing resource allocation, minimizing unnecessary setups, balancing reconfiguration time and cost with production needs, and improving capacity utilization, delivery reliability, downtime control, and operational continuity, although these benefits require empirical validation.

8 Limitation and Future Work

The manuscript focuses on framework design, required data flows, and decision logic rather than a detailed implementation or experimental deployment. The present study has the following limitations:

- a single illustrative case evaluating only three RPs
- use of deterministic inputs
- absence of physical deployment
- lack of validation of DT fidelity
- limited problem scale
- omission of human and organizational factors
- dependence on validated shop-floor data

The manuscript's contributions lie in the illustrated conceptual architecture that formalizes RP selection, integrates DT-based data flows, and structures managerial decision logic. Outcomes such as real-time DT synchronization**,** autonomous policy updates, and learning-enabled adaptation remain future implementation effects that require empirical testing. This now separates these two layers to clarify what is already established versus what must be demonstrated in practical deployment.

Future extensions of this work may include optimization of MPFP sequences for seasonal or time-varying part families and coupling the DT with simulation and optimization platforms such as Enterprise Dynamics, Tecnomatix Plant Simulation, and MATLAB to evaluate alternative schedules, validate reconfiguration policies, and identify globally optimal MPFP plans. The framework may be extended to cases in which the part-family production order is also a decision variable. In future, an implementation study in controlled experiments can be carried out that compare the DT-driven adaptive policy against three baseline rules for e.g., fixed production order, minimum-cost, and minimum-time using identical input data and constraints for every simulation run. Each policy can be evaluated on the same demand profiles, machine capabilities, and reconfiguration matrices, with performance measured by total reconfiguration time, total reconfiguration cost, and number of reconfiguration setups. This can be accompanied with sensitivity analyses including discrete two-level tests and continuous parameter sweeps by varying reconfiguration

cost and family demand to assess the robustness of policy selection and to identify conditions where the adaptive approach provides clear advantages.

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Data Availability

The data supporting the findings of this study are within the article.

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Conflicts of Interest

The author declares no conflicts of interest.

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Nomenclature

CPPS	Cyber-physical production systems
DT	Digital twin
DRL	Deep reinforcement learning
GKE	Generalized knowledge encapsulation
MPFP	Multiple Part Family Production
MQTT	Message Queuing Telemetry Transport
OPC-UA	Open Platform Communications-Unified Architecture
PFF	Part family formation
RMS	Reconfigurable manufacturing system
RFL	Reconfigurable flow line
RMT	Reconfigurable machine tool
RP	Reconfiguration policy
w_i	Workstation number on an RFL, $i \in \{1, 2, \ldots, m\}$
f_j	Part family to be produced on the RFL, $j \in \{1, 2, \ldots, 5\}$
T_k	RMT number, $k \in \{1, 2, 3, 4\}$
T_k^p	Potential RMT configurations, $\forall k \in \{1, 2, 3, 4\},\ p \in \{1, 2\}$
$O_l^{f_j}$	Operation 'l' to be performed on the jth part family f_j, $l \in \{1, 2, \ldots, 8\}$, $j \in \{1, 2, \ldots, 5\}$
$\tau_{k \to k'}^{p \to p'}$	Reconfiguration time required to shift between RMT configurations, min; $\forall\{k, k'\} \in \{1, 3, 4\},\ \{p, p'\} \in \{1, 2\};\ \forall\{k, k'\} \in \{2\},\ \{p, p'\} \in \{1\}$
$C_{k \to k'}^{p \to p'}$	Reconfiguration cost of switching between RMT configurations, 10^3 USD; $\forall\{k, k'\} \in \{1, 3, 4\},\ \{p, p'\} \in \{1, 2\};\ \forall\{k, k'\} \in \{2\},\ \{p, p'\} \in \{1\}$
$R_{w_i}^{O_l}$	Production rate of an RMT configuration for performing the lth operation on the w_ith stage of an RFL, parts/h